# Reusable Mathematical Models for Fusion Diagnostics

## Reduced plasma diagnostics and repeated control-law evaluations

Scientific working paper | Status: research proposal | 4 November 2026

### Research opportunity

Magnetic confinement fusion requires high-frequency interpretation of measurements and reliable evaluation of control quantities. While the underlying plasma dynamics are complex, selected diagnostic corrections, response surfaces and constraints can be expressed as bounded mathematical functions. paceval UG proposes studying such functions as reusable mathematical graphs that can eventually be considered for programmable hardware execution.




**Figure 1. Proposed integration boundary for a reusable mathematical evaluation kernel.**

### 1. Existing research tools and the exact gap to investigate

Tokamak experiments have demonstrated advanced plasma control strategies, including deep reinforcement learning. The peer-reviewed work of Degrave and colleagues is an example of a successful control system at the TCV facility. This does not imply that the complete plasma simulation or trained controller is an appropriate substitute workload for paceval. The proposed investigation concerns explicitly defined submodels selected by a domain expert. [1]

### 2. Research contribution proposed by paceval

For recurring diagnostic and actuator-limiting calculations, reusable graphs may allow efficient evaluation across different sensor readings and model parameters. Independent numerical checks may complement existing quality assurance. Any application in a real control loop must respect machine-specific latency, stability and operational constraints.

paceval is a mathematical engine that transforms supported text expressions into reusable computation objects. Its published C/C++ API supports preparing calculations for repeated evaluation, including batched input sets. The initial study should use local library integration. Networked REST or MCP communication would distort latency measurements for the targeted arithmetic kernel. The product documentation is linked in the references.

### 3. Explicit mathematical test case

An explicitly illustrative reduced model is y_hat = a_0 + a_1 I_p + a_2 B + a_3 I_p B + a_4 B^2. Here I_p is a chosen plasma-current signal and B is a specified magnetic-field observable. To avoid physical dimensional ambiguity, the plotted variables are normalized and all coefficients are dimensionless. This polynomial is a benchmark expression, not a validated model of an actual tokamak.

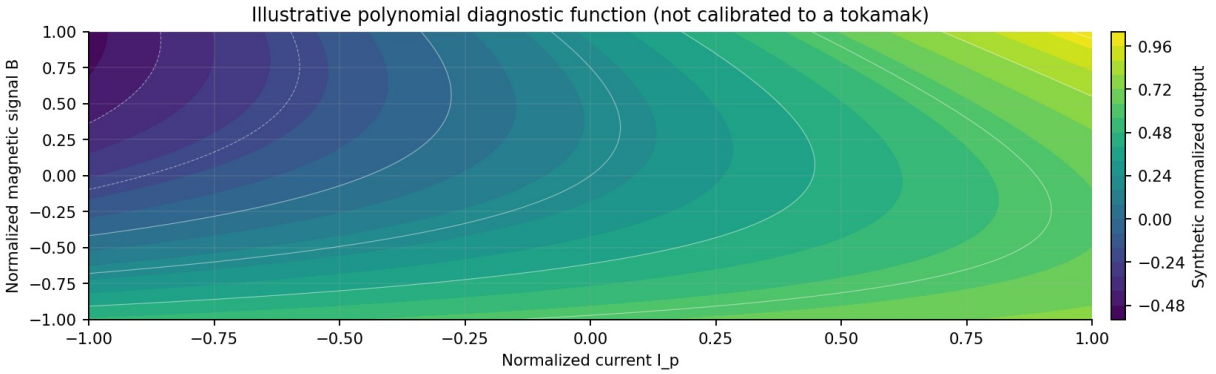



**Figure 2. Illustrative reduced diagnostic response surface. Synthetic or analytic example, not a paceval performance measurement.**

The contour plot illustrates a generic polynomial response surface with deliberately synthetic coefficients. It only demonstrates a family of repeated, parameterized arithmetic evaluations. Physical diagnostic usefulness must be established from experimental data by the collaborating fusion group.

## 4. Reproducible pilot experiment

With a fusion partner, choose one operationally meaningful explicit diagnostic or cost function. Define sensor units, normalization, valid operating envelope, noise assumptions and out-of-range behavior. Evaluate the same function in optimized C++ and paceval over recorded or synthetic values. Report both mathematical accuracy and any reduction in total diagnostic time, not only a microbenchmark. Validate uncertain values with independent references.

| Evaluation dimension | Study design |
|---|---|
| Mathematical correctness | Identical formulas and input domain. Independent numerical references, including boundary cases. |
| Kernel efficiency | Evaluations per second; end-to-end diagnostic latency; maximum observed latency; validity against physical reference data; numerical deviations; memory, watts and FPGA resource estimates. |
| Reproducibility | Thirty or more repetitions after warm-up, published inputs and software versions. Report median, dispersion and uncertainty. |
| Full-system validity | Include graph creation, adapters, data movement and the effect on total execution time. |

## 5. Integration with established scientific software

Place the optional evaluation kernel after sensor preprocessing, returning computed quantities and error status to an existing supervisory controller. Keep the safety loop, actuator commands and any real-time scheduling under the established facility control software. First test completely offline, then in a shadow-mode experiment if the research partner approves.

## 6. Programmable FPGA and ASIC perspective

A programmable FPGA mathematical path is attractive for fixed-rate, repeated expression graphs. Its possible benefit depends on the measurement of data transfers, fixed latency, numerical equivalence, resource utilization and power. Hardware suitability is an open engineering result.

## 7. Expected scientific contribution

This pilot could create a transparent methodology for isolating, validating and accelerating small but important mathematical functions in fusion diagnostics without modifying an existing plasma control architecture.

## 8. Open invitation to collaborate

paceval UG welcomes collaborations with fusion laboratories that can identify a valuable explicit kernel. Together, the partners could develop a reproducible study and refine the API for robust mathematical execution in scientific control workflows.

paceval UG is prepared to make the existing mathematical technology available for jointly defined research and to discuss targeted API extensions, open benchmark artifacts and integration formats with the scientific partners. Publication rights, licensing and intellectual property would be agreed before a joint project begins.

## 9. Practical work packages

First, identify a diagnostic expression and a valid operating envelope with a fusion laboratory. Next, perform offline kernel verification on recorded or synthetic inputs. Finally, test a shadow-mode adapter and separately design future hardware experiments if the facility permits.

## Scientific and technical sources

[1] Degrave et al., magnetic control of tokamak plasmas, Nature 602 (2022). Open source 1

[2] acados documentation for control workflows. Open source 2

[3] paceval C/C++ API. Open source 3

[4] FLINT / Arb interval arithmetic documentation. Open source 4

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